

When Wholesalers Join the Cartel: Vertical Coordination in Gasoline Markets

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Abstract

This paper studies how upstream participation changes cartel organization and outcomes in Brazilian gasoline markets. Using antitrust case records, I document that vertical cartels involve wholesalers in coordinating retailers and disciplining deviations through wholesale terms, unlike retail cartels relying on downstream coordination. I combine eight cartel cases with station-level price data and a matching difference-in-differences design, finding larger retail-price and retail-markup effects in vertical-cartel cases, while small retailers capture smaller markup gains and face higher wholesale markups. A model shows how discriminatory wholesale terms weaken retailers' incentives to remain outside the cartel, broaden participation, and support higher retail prices.

Keywords: Collusion, Vertical Contracting, Fuel Retail, Antitrust Enforcement, Nash Bargaining

JEL Codes: L41, L42, K21, L71, C23

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1 Introduction

Vertical integration and vertical coordination have long occupied an ambiguous place in antitrust analysis. Classic contributions emphasize that closer alignment between upstream and downstream firms may improve pricing efficiency and, in some environments, reduce final prices (Spengler, 1950; Rey and Tirole, 2007). By eliminating successive markups or facilitating coordination along the supply chain, vertical relationships are often viewed as benign or even welfare-enhancing relative to horizontal concentration.

At the same time, vertical arrangements can generate competitive harm by foreclosing rivals, softening downstream competition, or facilitating collusion. More recently, theoretical and empirical research has shown that vertical links can play a central role in sustaining collusion, particularly in hub-and-spoke settings in which upstream firms help coordinate or discipline downstream rivals (Jullien and Rey, 2007; Van Cayseele and Miegielsen, 2013; Harrington, 2017; Clark et al., 2024; Chaves and Duarte, 2025).

What remains less understood is how vertical participation changes the *magnitude* and *organization* of collusive outcomes. The literature has largely focused on whether vertical relationships make collusion easier to sustain. Less is known about whether upstream participation makes cartels more harmful once collusion occurs. Do vertical cartels generate larger retail-price effects than purely horizontal retail cartels? Do they affect wholesale margins? And do they change which downstream firms participate in the cartel?

This paper answers these questions using evidence from Brazilian gasoline retail markets. I distinguish between *retail-horizontal cartels*, which I refer to hereafter as horizontal cartels, and *vertical cartels*. I classify a case as horizontal when the legal records document cartel participation by retailers but provide no evidence that wholesalers either formed a cartel themselves or actively assisted retailers in organizing, monitoring, or enforcing the downstream agreement. By contrast, I classify a case as vertical when the records provide evidence that retailers and wholesalers participated jointly in the collusive arrangement, with

wholesalers actively supporting downstream coordination.

The documentary evidence provides direct information on how these two types of cartels were organized and enforced. I use public records from the Brazilian Antitrust Authority (CADE) covering eight gasoline cartel cases, four horizontal and four vertical. In horizontal cases, coordination and enforcement relied primarily on retailers, which communicated about prices, monitored posted prices, and pressured stations that deviated from the agreement. In vertical cases, wholesalers or their representatives also participated in price and margin discussions and could use their commercial relationships with stations to support downstream coordination. In the clearest cases, the records document the use of wholesale prices, discounts, financial offsets, and other commercial terms to reward compliant retailers or worsen the conditions faced by stations that deviated from coordinated pricing. These records therefore identify an additional enforcement technology available to vertically organized cartels through the wholesale relationship.

I combine this documentary evidence with a station-level panel of weekly retail and wholesale prices and station characteristics to examine whether cartels with active wholesaler participation also generate systematically different market outcomes. I estimate the effects of cartel dissolution using a matching difference-in-differences design that compares cartel-affected markets with similar nearby control markets and treated stations with observationally similar stations in those controls. Vertical cases exhibit substantially larger effects on retail prices and retail markups than horizontal cases, while the evidence for wholesale markups is less precise but points in the same direction. At the station level, small retailers under vertical cartels experience similar retail-price effects but smaller retail-markup gains than larger retailers while facing higher wholesale markups. This heterogeneity is consistent with the documentary evidence that wholesale terms could be used to discipline retailers with stronger incentives to deviate or remain outside the agreement. Because wholesaler participation is not randomly assigned across cartel cases, the reduced-form comparison is not intended to isolate the effect of upstream participation independently of all other un-

observed differences across cartels. Instead, it complements the documentary evidence by showing that cases in which the additional upstream enforcement mechanisms are observed are also associated with substantially larger retail-price and markup effects.

I rationalize these patterns with a model of capacity-constrained retail collusion with upstream participation, building on Bos and Harrington (2010). In a horizontal retail cartel, some retailers may prefer to remain outside the agreement and form a competitive fringe. Fringe capacity limits the residual demand available to cartel members and constrains the retail price that the cartel can sustain. When wholesalers participate in the agreement, they can instead discriminate between cartel participants and outsiders through wholesale terms. By worsening the wholesale conditions faced by non-participants, vertical coordination weakens the attractiveness of remaining outside the cartel, supports broader downstream participation, and allows the cartel to sustain higher retail prices. The insider wholesale price then determines how the additional surplus generated by vertical coordination is divided between wholesalers and retailers. A calibrated numerical exercise in the Appendix shows that these mechanisms can reproduce the qualitative ranking of cartel effects observed in the data.

The paper contributes to three literatures. First, it contributes to the empirical literature on cartels and antitrust interventions by showing that the vertical organization of collusion matters for both the magnitude of overcharges and the distribution of cartel gains across firms (Genesove and Mullin, 2001; Asker, 2010; Byrne et al., 2025). The paper is particularly related to Holler and Rickert (2022), who show how resale price maintenance can strengthen upstream collusion by aligning retailer incentives. I study a complementary mechanism in which wholesalers actively participate in downstream collusion and use wholesale terms to discipline retailers. Rather than focusing on whether vertical restraints increase cartel stability or duration, I estimate how upstream participation changes the effects of collusion on prices and margins.

Second, the paper contributes to the literature on vertical restraints and hub-and-spoke collusion by showing that upstream participation can do more than facilitate coordination:

it can change downstream cartel participation and amplify collusive harm (Jullien and Rey, 2007; Van Cayseele and Miegielsen, 2013; Harrington, 2017; Hatfield et al., 2020; Clark et al., 2024; Chaves and Duarte, 2025; Gilo and Yehezkel, 2025). Relative to Clark et al. (2024), who emphasize supplier-retailer contracting frictions and asymmetries generated by main-supplier relationships, my mechanism operates through capacity constraints, fringe participation, and discriminatory wholesale prices. The paper is also closely related to Chaves and Duarte (2025), who study a single Brazilian gasoline cartel and emphasize incomplete cost pass-through under exclusive supply agreements. I instead study eight cartel cases and exploit variation between horizontal and vertical cartels to examine how wholesalers' active participation changes cartel outcomes.

Third, the paper connects cartel organization to the literature on vertical bargaining by showing how wholesale terms can simultaneously discipline downstream participation and allocate collusive surplus (Horn and Wolinsky, 1988; Berto Villas-Boas, 2007; Grennan, 2013; Crawford et al., 2018; Collard-Wexler et al., 2019). Existing models of bargaining over cartel profits primarily study surplus division among heterogeneous firms within a horizontal cartel (Saglam, 2018; Hattori, 2021). In contrast, I study bargaining across the upstream and downstream tiers of a vertically organized cartel. To the best of my knowledge, this is the first paper to model the division of hub-and-spoke cartel profits using a Nash bargaining framework.

These findings have direct implications for antitrust enforcement. Treating cartels as a homogeneous category may understate harm when upstream firms can weaken fringe competition, induce broader downstream participation, and discipline retailers through wholesale terms. Recognizing vertical participation as a distinct dimension of cartel organization is therefore important for measuring damages, designing remedies, and prioritizing enforcement.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting, data, and cartel cases. Section 3 presents documentary evidence on cartel organiza-

tion and enforcement mechanisms. Section 4 presents the reduced-form evidence comparing horizontal and vertical cartels. Section 5 develops the theoretical model, and Section 6 concludes. Additional matching details, theoretical derivations and proofs, calibration exercises, and robustness checks are reported in the Appendix and Online Appendix.

2 Institutional Background and Data

2.1 Brazilian Gasoline Market and Cartel Cases

This paper studies the Brazilian gasoline retail market. Gasoline markets have been widely analyzed in the collusion literature because they combine frequent interaction, relatively homogeneous products, and publicly observable retail prices (Clark and Houde, 2013, 2014; Byrne and De Roos, 2019; Chaves and Duarte, 2025). The Brazilian market shares these features. Fuel distributors, which I refer to as wholesalers, supply gasoline to gas stations, which then sell it to final consumers. Gas stations are also required to publicly display their retail prices, making competitors' pricing decisions readily observable.

The Brazilian market has several additional institutional features that are particularly relevant for studying vertical participation in cartels. Retailers are predominantly local operators, with gas-station networks typically concentrated within a single city or metropolitan area, so downstream competition is largely local. By contrast, the wholesale market consists of a relatively small number of national firms that operate across many regions and delegate local commercial decisions and relationships with stations to regional managers. In addition, formal vertical integration is legally prohibited: fuel distributors cannot own or operate gas stations. Relationships between wholesalers and retailers therefore arise through contractual arrangements rather than common ownership.

Cartel investigations are conducted by law-enforcement authorities in coordination with the Brazilian Antitrust Authority (CADE). To establish cartel liability, CADE relies on legal

evidence of coordination among firms; economic evidence, such as unusual price movements or high margins, is not sufficient on its own to establish that a cartel occurred. Investigations therefore rely on evidence such as leniency agreements, wiretaps, seized documents, information from regulatory agencies, and third-party complaints. I use publicly available CADE records, which provide information on the geographic scope and duration of the cartels, participating firms, and the involvement of retailers, wholesalers, and their local representatives.

The analysis focuses on eight gasoline cartel cases considered by CADE between 2004 and 2020. I include cases that either resulted in a cartel conviction or for which the administrative record contained evidence of cartel conduct but liability was ultimately not imposed because of procedural or legal rulings that prevented CADE from relying on that evidence. Thus, dismissal on procedural grounds is not interpreted as evidence that the underlying conduct did not occur. For example, in Case 1, the evidence collected during the investigation was subsequently invalidated after a court ruled that the underlying police operation was unlawful. Because CADE can establish cartel liability only on the basis of legally admissible evidence, the case could no longer support a conviction even though the case file contained documentary and economic evidence consistent with collusion. I further restrict the sample to cases with a documented or alleged cartel duration of at least two months, ensuring a sufficiently long pre-dissolution period for the empirical analysis.

Within this sample, I distinguish between *retail-horizontal cartels*, hereafter referred to as horizontal cartels, and *vertical cartels*. A case is classified as horizontal when the legal records document cartel participation by retailers but provide no evidence that wholesalers either formed a cartel themselves or actively assisted retailers in organizing, monitoring, or enforcing the downstream agreement. A case is classified as vertical when the records provide evidence that retailers and wholesalers participated jointly in the collusive arrangement, with wholesalers or their regional representatives actively supporting downstream coordination. The resulting sample contains four horizontal and four vertical cases.

Cartel dates are determined from these legal records rather than from observed price movements. The start date corresponds to the earliest documented evidence of collusive conduct, whereas the end date is generally associated with the disruption or public disclosure of the cartel investigation, such as arrests or search-and-seizure operations. The resulting treatment period should therefore be interpreted as the *legally documented cartel period*. In this sense, the dating is conservative: during the identified cartel period, there is direct legal evidence indicating that the collusive agreement was operating, while collusion may have begun before its first documented occurrence. An additional advantage is that the cartel period is defined independently of the price outcomes analyzed below, avoiding a mechanical relationship between treatment timing and the dependent variables. Table 1 summarizes the geographic scope, duration, and classification of the eight cases.

Table 1: Cartel Dating

Case Number	Case Legal Code	Cartel Start	Cartel End	Relevant Market	Structure
1	08700.003447/2015-40	Jul/01	May/07	Cuiabá/MT, Várzea Grande/MT	Vertical
2	08012.010215/2007-96	Jul/04	Apr/06	Caxias do Sul/RS	Horizontal
3	08012.007866/2007-07	Jan/06	Apr/07	João Pessoa/PB	Vertical
4	08012.008847/2006-17	Jul/06	Mar/07	Vitória/ES, Cariacica/ES, Vila Velha/ES, Serra/ES	Horizontal
5	08012.011668/2007-30	Jan/07	Aug/07	Cambé/PR, Londrina/PR	Horizontal
6	08700.010769/2014-64	Feb/07	May/08	Belo Horizonte/MG, Betim/MG, Contagem/MG	Vertical
7	08700.000547/2008-95	Sep/08	Dec/09	Teresina/PI	Horizontal
8	08012.008859/2009-86	Jan/11	Feb/16	Brasília/DF	Vertical

Note: This table refers to the dates defined by each legal proceeding. The Legal Code refers to the number attributed by the Antitrust Authority to that cartel case and contains all its documents. Cartel Start and Cartel End refer to the cartel period, while relevant market refers to the cities affected by the cartel.

2.2 Price Data, Sample, and Outcomes

The main data come from the weekly Fuel Price Panel collected by the Brazilian Oil, Natural Gas, and Biofuels Agency (ANP), which reports retail gasoline prices charged by individual gas stations and the wholesale prices paid by stations to fuel distributors from May 2004 to August 2020. The survey is not designed as a balanced panel: stations are sampled to maximize geographic coverage rather than according to a fixed establishment-level sampling rule (Silveira et al., 2022). Consequently, the composition of observed stations varies over time. I also use regional gasoline production and import prices published by ANP as a proxy for wholesalers’ marginal costs. These measures vary across regions and over time and capture the main input-cost component faced by wholesalers.

The main outcomes are retail prices, retail markups, wholesale prices, and wholesale markups. Retail and wholesale prices are directly observed in the ANP data. I define the retail markup and wholesale markup of station i in week t as

$$\text{Retail Markup}_{it} = \frac{p_{rit} - p_{wit}}{p_{rit}}, \quad \text{Wholesale Markup}_{it} = \frac{p_{wit} - c_{jt}}{p_{wit}},$$

where p_{rit} is the retail price charged by the station, p_{wit} is the wholesale price it pays and c_{jt} is the regional gasoline production and import cost in region j and week t .

I combine the ANP data with the CADE records to construct a station-week panel around each cartel episode. Treated stations are located in cities identified by CADE as part of the cartel’s relevant market during the legally documented cartel period. To construct the pool of potential controls, I use cities located 50–200 kilometers from the affected market. This choice follows CADE’s own practice when evaluating the economic evidence associated with gasoline cartels where it compares prices and margins in the relevant market with those observed in nearby cities, typically within a radius of up to 200 kilometers. This provides a useful basis for defining the comparison set because the relevant-market classification itself is informed by the documentary and economic evidence collected during the investigation.

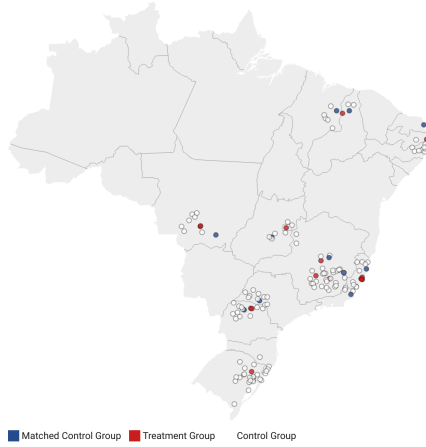
If a nearby city were found to be directly affected by the cartel, it would generally be incorporated into the affected market rather than treated as an unaffected comparison city. I exclude cities located within 50 kilometers of the relevant market to reduce the risk of spillover effects from cartel conduct or from the enforcement intervention.

To construct a more comparable control group, I adopt a matching strategy that selects control markets and stations based on observable demand- and supply-side characteristics; the procedure is described in detail in the next section. For this purpose, I supplement the price data with station characteristics from PostoWeb, including gasoline storage capacity, number of pumps, and exclusive supply status, and with population and vehicle ownership from the Brazilian Institute of Geography and Statistics (IBGE). I use station storage capacity to construct a retailer HHI for each market, with each station network’s market share measured by its share of total gasoline storage capacity. I also construct a wholesaler HHI based on stations’ exclusive supply relationships.¹ The resulting dataset contains 7,644 gas stations across 156 cities. Figure 1 shows the geographic distribution of cartel-affected and potential control cities, while Table 2 reports summary statistics.

An additional concern is that vertical and horizontal cartels may arise in systematically different markets. Table 3 compares observable characteristics of cities affected by each cartel type. The main differences are largely driven by Brasília and Belo Horizonte, two exceptionally large markets relative to the other cartel cases. As shown in the final column of Table 3, excluding these two cities makes the observable characteristics of markets affected by vertical and horizontal cartels substantially more similar. In the baseline empirical analysis, Belo Horizonte is excluded because the matching procedure described in Section 4.1 does not identify a sufficiently comparable control market, whereas Brasília remains in the estimation sample. To assess whether the results are disproportionately influenced by Brasília, Online Appendix D re-estimates the main specifications after additionally excluding Brasília. The

¹For the wholesaler HHI, stations with an exclusive supply agreement are assigned to the corresponding wholesaler. Each station without an exclusive supply agreement is treated as a separate independent supplier in the calculation of market shares.

Figure 1: Cartel and Control Cities



Note: The figure plots cartel-affected cities, potential control cities, and matched control cities. Treated stations are located in cities identified by CADE as affected by cartel conduct. Potential control stations are located in cities between 50 and 200 kilometers from the affected market; cities within 50 kilometers are excluded to reduce potential spillover effects.

main findings remain qualitatively unchanged.

3 Documentary Evidence on Cartel Organization

Before turning to the reduced-form evidence, I use CADE’s administrative records to characterize how the cartels operated. The purpose of this section is not to reassess legal liability, but to identify the organizational mechanisms documented in the cases. The central distinction is that horizontal cartels relied primarily on coordination and enforcement among retailers, whereas vertical cartels could additionally use wholesalers and the wholesale relationship to monitor retailers, discipline deviations, and allocate margins along the supply chain.

In the horizontal cases, coordination occurred at the retail level. Retailers used meetings, phone calls, trade associations, reference stations, and local leaders to agree on prices and monitor compliance. The Vitória/ES case provides a clear example: station owners repeatedly communicated about the retail prices to be charged and used coded expressions such as

Table 2: Summary Statistics

	All Cities	Relevant Market	Matched Cities
Station-level Data			
Retail Price (BRL)	2.78 (0.592)	2.73 (0.596)	2.83 (0.636)
Wholesale Price (BRL)	2.41 (0.540)	2.41 (0.541)	2.45 (0.573)
Wholesale Cost (BRL)	1.63 (0.340)	1.65 (0.341)	1.66 (0.346)
Retail Markup	0.13 (0.038)	0.12 (0.042)	0.13 (0.036)
Wholesale Markup	0.47 (0.088)	0.46 (0.087)	0.48 (0.094)
Observations	902,854	261,507	139,960
City-level Data			
Retailer HHI (2024)	1,204 (1,486)	317 (240)	418 (411)
Wholesaler HHI (2024)	2,023 (1,524)	1,586 (623)	1,329 (724)
Vehicles (2024)	78,823 (200,037)	362,738 (519,144)	147,292 (180,855)
Population (2024)	213,067 (384,151)	815,219 (799,909)	401,542 (398,884)
Observations	156	15	11

Note: Entries report means, with standard deviations in parentheses. Relevant Market refers to cartel-affected cities. Matched Cities refers to the matched control markets described in Section 4.1.

Table 3: Market Characteristics by Cartel Type

	Horizontal	Vertical	Vertical excluding Brasília and Belo Horizonte
Retailer HHI	350.6 (305.0)	279.1 (153.7)	298.1 (126.7)
Wholesaler HHI	1,454.5 (580.8)	1,736.9 (681.4)	1,457.5 (595.5)
Vehicles	160,970 (73,956)	593,332 (711,450)	182,364 (64,898)
Population	482,961 (227,712)	1,194,944 (1,056,923)	593,090 (225,317)
Cities	8	7	5

Note: Entries report means, with standard deviations in parentheses. The final column excludes Brasília/DF and Belo Horizonte/MG from the vertical-cartel sample.

“math,” “engine,” and “machine configuration” when discussing price changes. The records show that retailers monitored competitors’ posted prices and adjusted their own prices in response. Similar leader–follower structures appear in Caxias do Sul/RS and Londrina/PR, where a small number of large or strategically located stations acted as reference points for coordinated price changes. Such arrangements resemble those documented in other gasoline markets (Clark and Houde, 2014; Byrne and De Roos, 2019).

Horizontal coordination did not require every retailer to participate. The records describe stations that did not initially follow coordinated price increases and groups of cartel members responsible for visiting or pressuring them. Enforcement therefore remained pri-

marily downstream, with retailers monitoring posted prices, communicating directly with one another, and using peer pressure or threats to discipline deviations. This feature is important for the model below because retailers that remain outside the agreement can constrain the price sustained by a purely horizontal cartel.

Vertical cartels differed from horizontal cartels because wholesalers were active participants in the collusive arrangement rather than passive observers of downstream coordination. The administrative records document wholesaler representatives participating in meetings and communications with retailers, helping organize the agreement, and intervening in the relationship between the cartel and individual stations. Their involvement therefore expanded the scope of coordination from the retail level to the vertical relationship between wholesalers and retailers.

The first mechanism concerns the joint determination of prices and the allocation of cartel rents across the supply chain. In the vertical cases, wholesalers participated in discussions over the retail prices to be charged while also influencing the wholesale prices and margins faced by stations. The Cuiabá/MT record, for example, documents discussions over margins and the division of the retail margin between distribution and retail. This evidence suggests that retail and wholesale terms were not determined independently. Instead, vertical coordination allowed the two tiers to jointly determine the final retail price and use wholesale terms to divide the resulting cartel surplus between wholesalers and retailers. This feature motivates the model's treatment of the vertical cartel as a joint arrangement rather than as independent upstream and downstream cartels.

The second mechanism is discriminatory wholesale pricing used to discipline downstream participation. The Belo Horizonte/MG records state that wholesalers provided financial offsets in the amounts charged to retailers when stations changed retail prices according to their instructions, and that these benefits could be withdrawn depending on the station's behavior. A retailer that deviated from the coordinated agreement could therefore face not only punishment from other retailers but also less favorable wholesale terms. By changing

the payoff from deviation or non-participation, wholesalers could weaken retailers' outside options and make broader downstream participation easier to sustain. This evidence provides the documentary basis for the discriminatory wholesale-price mechanism developed in the model.

The Brasília/DF case is treated more cautiously. The retail cartel is strongly documented, but the distributor-related allegations were ultimately archived for insufficient evidence. The case file nevertheless contains allegations concerning buyer discrimination, the promotion of uniform commercial conduct, and market division. I therefore do not use Brasília as the strongest evidence of wholesaler liability, but retain it as informative about the forms of vertical conduct considered during the investigation.

Table 4 summarizes the mechanisms documented across the eight cases. The main distinction is that horizontal cartels relied on downstream communication, monitoring, and pressure, whereas vertical cartels had access to an additional enforcement technology through the wholesale relationship. The empirical analysis below asks whether cartels with this additional upstream channel generate larger effects on retail prices, retail markups, and wholesale markups than purely horizontal retail cartels. Additional case-specific evidence on cartel organization and enforcement mechanisms is provided in Online Appendix B.

4 Horizontal vs. Vertical Cartels

This section compares the effects of horizontal and vertical cartels on retail prices, retail markups, and wholesale markups. I first estimate the city-level effects of cartel dissolution and test whether they differ by cartel structure. I then use matched station-level panels to examine heterogeneity by station size, which is central to the mechanism developed in the theoretical model. Additional details on the matching procedure are provided in Online Appendix A, while robustness checks for the empirical results are reported in Online Appendix D.

Table 4: Documented Cartel Mechanisms by Case

Case	Type	Participants	Communication	Assoc.	Leader	Retail mon.	Retail pres.	Whol. reps.	Whol. pres.	Whol. terms
Case 1	V	Retailers, retail association, wholesaler representatives	Association meetings and direct contacts among retailers and upstream agents	Yes	Partial	Yes	Yes	Yes	Partial	Yes
Case 2	H	Retailers and large retail networks	Phone calls, meetings, and direct coordination among retailers	No	Yes	Yes	Yes	No	No	No
Case 3	V	Retailers, retail association, wholesaler representatives	Association meetings, phone calls, and contacts involving upstream agents	Yes	Partial	Yes	Yes	Yes	Partial	Partial
Case 4	H	Retailers	Phone calls, coded language, and meetings	No	No	Yes	Partial	No	No	No
Case 5	H	Retailers and reference stations	Phone calls and coordinated timing of price changes	No	Yes	Yes	Yes	No	No	No
Case 6	V	Retailers, retail association, reference stations, wholesaler representatives	Association contacts, direct calls, and vertical coordination between retailers and upstream agents	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Case 7	H	Retailers and retail association	Union meetings, meeting minutes, and direct statements	Yes	No	Yes	Yes	No	No	No
Case 8	V	Retailers, large networks, wholesaler representatives	Direct contacts, network coordination, and intercepted communications	Partial	Yes	Yes	Yes	Partial	Partial	Partial

Note: H denotes horizontal cartels and V denotes cartels with vertical participation. “Assoc.” indicates participation of a retail association or union in the organization of the cartel. “Leader” indicates the presence of leader firms or reference stations. “Retail mon.” indicates monitoring of posted retail prices. “Retail press.” indicates pressure, threats, visits, or other downstream enforcement against deviating retailers. “Whol. reps.” indicates participation, accusation, or documented involvement of wholesalers’ local employees, managers, or commercial representatives. “Whol. press.” indicates pressure through the wholesale relationship, brand relationship, or supply network. “Whol. terms” indicates evidence that upstream agents could influence discounts, effective wholesale prices, financial offsets, or the division of margins between wholesale and retail. “Partial” indicates that the mechanism appears in the record but is weaker, indirect, alleged, archived, or not central to the final legal characterization of the case.

4.1 Matching Strategy

As shown in Table 2, cartel-affected markets differ substantially from the full set of potential control markets along several observable characteristics. Directly comparing treated markets with all unaffected cities may therefore generate a poor counterfactual. To improve comparability, I adopt a two-step matching strategy. This approach is closely related to exact-matching designs used in the labor literature (Jäger and Heining, 2022; Goldschmidt and Schmieder, 2017; Smith et al., 2019). Among these applications, the closest to my setting is Arnold (2019), who uses a sequential matching procedure that first matches firms and then matches workers within those firms. I adopt an analogous two-step structure by first matching markets on observable characteristics and then matching gas stations within the selected markets. To the best of my knowledge, this paper is the first to apply this type of sequential matching design to the estimation of cartel effects.

In the first stage, I match each cartel-affected city to a nearby control city using retailer concentration, wholesaler concentration, population, and the number of registered vehicles. Because historical concentration measures are not consistently available, I use the most recent available measures, from 2024. The covariates are standardized and matched using nearest-neighbor Euclidean distance with replacement. This stage is intended to select control markets that are similar to treated markets in demand conditions and in the structure of both the retail and wholesale sectors. All cartel-affected cities obtain a sufficiently close match except Belo Horizonte, which is therefore excluded from the baseline empirical analysis. Online Appendix A.1 provides additional details and reports the resulting matches.

In the second stage, I construct a matched station-level sample within each treated city and its selected control market. Using Coarsened Exact Matching, I match treated and control stations observed in the same calendar week on gasoline storage capacity, number of pumps, and exclusive supply status. This second stage improves comparability at the establishment level after conditioning on market-level similarity. Approximately 74 percent of

treated station-week observations are retained in the matched sample. Online Appendix A.2 describes the station-level procedure in detail.

4.2 City-Level Effects

The empirical analysis uses cartel dissolution as the treatment event. The intuition is that, while a cartel is active, the affected market is exposed to collusive conduct; dissolution removes this conduct. I therefore compare changes in outcomes in the cartel-affected city around dissolution with contemporaneous changes in its matched control market. A decline in an outcome following dissolution implies that the cartel had raised that outcome while active.

The comparison between vertical and horizontal cases should not be interpreted as arising from random assignment of wholesaler participation. Unobserved characteristics of the cartel or market may be correlated with both cartel organization and its economic effects. The reduced-form estimates therefore complement, rather than replace, the documentary evidence. The legal records provide direct evidence on the mechanisms associated with upstream participation, while the empirical analysis examines whether cases exhibiting these mechanisms are also associated with systematically different effects on prices and markups.

I first examine the dynamics around dissolution using an event-study specification that allows the effects to differ between horizontal and vertical cartels:

$$Y_{mct} = \sum_{\substack{k=-10 \\ k \neq -1}}^{20} \lambda_k^H D_{mct}^k (1 - \text{Vertical}_c) + \sum_{\substack{k=-10 \\ k \neq -1}}^{20} \lambda_k^V D_{mct}^k \text{Vertical}_c + \psi_{c,t} + \alpha_m + \varepsilon_{mct}, \quad (1)$$

where Y_{mct} is the average log retail price, retail markup, or wholesale markup in city m , cartel case c , and week t . The indicator D_{mct}^k equals one for the cartel-affected city k weeks relative to dissolution. The specification includes city fixed effects, α_m , and case-by-calendar-week fixed effects, $\psi_{c,t}$, so identification comes from changes in the treated city relative to its

matched control within each cartel case. The omitted week is $k = -1$, and standard errors are clustered at the city level.

Figure 2 reports the estimates separately for horizontal and vertical cartels. Before dissolution, the coefficients fluctuate around zero for both cartel structures. Because cartels are already active during this period, these coefficients are not conventional pre-treatment tests; rather, they show that treated and matched control markets were not evolving differentially immediately before dissolution. After dissolution, retail prices and retail markups fall for both types of cartels, but the declines are larger for vertical cartels. Wholesale markups exhibit little change following horizontal-cartel dissolution but decline following the dissolution of vertical cartels. The dynamic evidence therefore anticipates the main cross-cartel comparison, with vertical-cartel cases exhibiting larger estimated effects, particularly in retail markets.

To summarize these effects, I estimate a difference-in-differences specification using a symmetric window around dissolution. I retain up to 78 weeks of cartel activity, approximately 18 months, and use a post-dissolution period of equal length for each case. For cartels lasting fewer than 78 weeks, I retain the full documented cartel period and an equally long post-dissolution window.

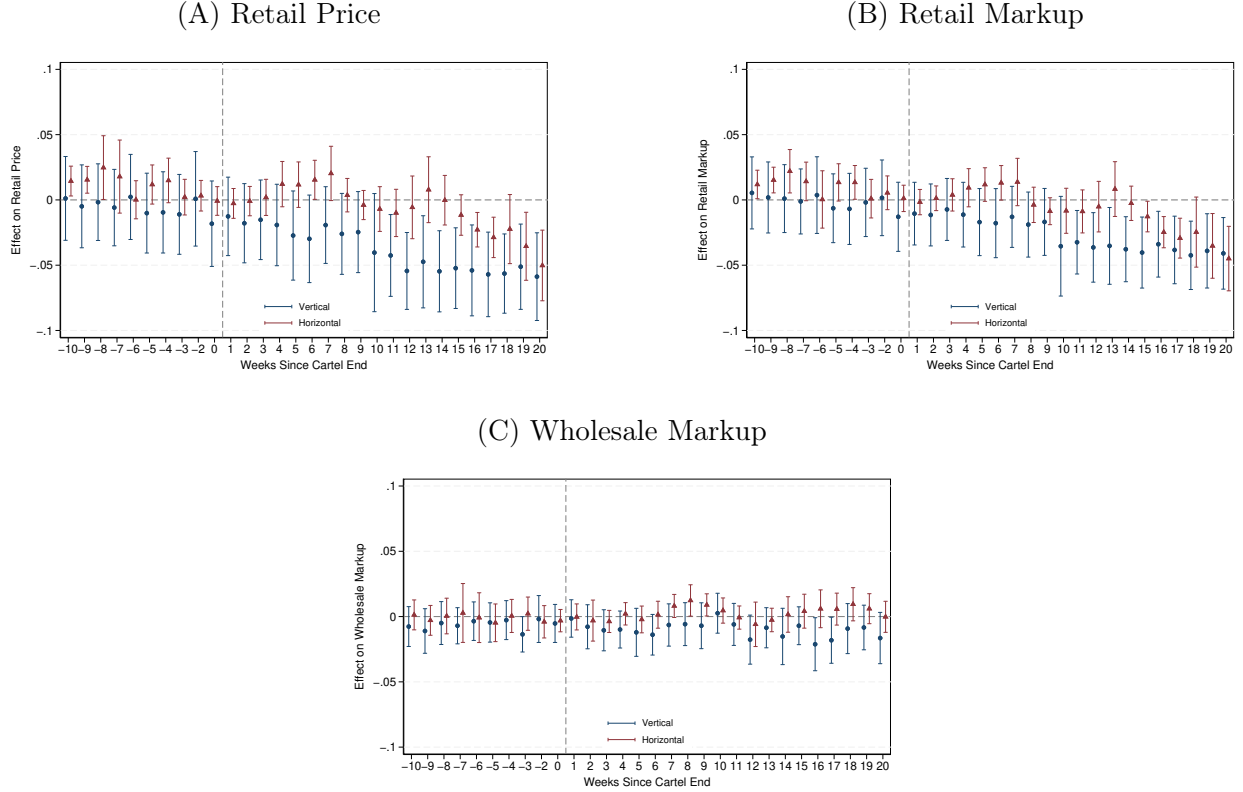
Using this sample, I estimate

$$Y_{mct} = \alpha_m + \psi_{c,t} + \beta_1 \text{Cartel}_{mct} + \beta_2 (\text{Cartel}_{mct} \times \text{Vertical}_c) + \xi_{mct}, \quad (2)$$

where Cartel_{mct} equals one for the affected city while the cartel is active and zero otherwise. Thus, β_1 measures the change in the outcome following the dissolution of a horizontal cartel, while β_2 captures the additional post-dissolution change associated with vertical cartels. Because the omitted period corresponds to cartel operation, the implied effect of a horizontal cartel while active is $-\beta_1$, whereas the implied effect of a vertical cartel is $-(\beta_1 + \beta_2)$.

Table 5 confirms the patterns in the event studies. Horizontal-cartel cases are associated

Figure 2: City-Level Event Studies by Cartel Structure



Note: Dots report the event-time coefficients for each cartel structure from equation (1). The omitted category is the week immediately preceding cartel dissolution. Vertical bars represent 95% confidence intervals. Time 0 corresponds to the week of cartel dissolution. Observations are weighted by the frequency with which each control city is used in the city-level matching procedure.

with an increase in retail prices of approximately 1.2%, while the estimated effect in vertical-cartel cases is approximately 4.4%, more than three times larger. Horizontal cartels increase retail markups by approximately 1.1 p.p., with vertical-cartel cases exhibiting an additional effect of approximately 2.6 p.p. For wholesale markups, the horizontal-cartel effect is close to zero, whereas the estimated effect in vertical-cartel cases is positive, although less precisely estimated. Taken together with the documentary evidence, the larger retail-price and retail-markup effects in vertical-cartel cases are consistent with the mechanisms documented in the legal records, whereby upstream participation provides additional instruments to discipline retailers and sustain downstream coordination. The positive wholesale-markup effect is also consistent with part of the additional collusive surplus being captured by wholesalers through

Table 5: City-Level: Effect of Cartel Dissolution

	(1)	(2)	(3)
	log(Retail Price)	Retail Markup	Wholesale Markup
Cartel	-0.012*** (0.004)	-0.011*** (0.002)	-0.000 (0.002)
Cartel $\times$ Vertical	-0.032*** (0.010)	-0.015** (0.007)	-0.021* (0.011)
Constant	0.986*** (0.001)	0.143*** (0.001)	0.459*** (0.000)
Case-Week Fixed Effects	Yes	Yes	Yes
City Fixed Effects	Yes	Yes	Yes
Observations	3,399	3,399	3,399

Note: This table reports estimates from specification (2). All regressions include case-by-calendar-week fixed effects and city fixed effects. Standard errors are clustered at the city level and are reported in parentheses. The omitted period corresponds to the period in which the cartel was operating; therefore, negative post-dissolution coefficients imply higher outcomes during cartel operation. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

the wholesale relationship.

Because identification of the difference between cartel structures relies on only eight cartel cases, I complement conventional clustered inference with exact permutation inference. With four horizontal and four vertical cases, there are only $\binom{8}{4} = 70$ possible assignments of cartel structure. I re-estimate equation (2) under each assignment and compare the observed interaction coefficient with the resulting permutation distribution. The observed coefficient ranks as the third most negative for retail prices, the sixth most negative for retail markups, and the fourteenth most negative for wholesale markups, corresponding to exact one-sided p -values of 0.043, 0.086, and 0.200, respectively. The exact inference therefore supports a larger effect of vertical cartels on retail prices, provides suggestive evidence for retail markups, and yields weaker evidence for wholesale markups.

4.3 Firm-Level Heterogeneity by Station Size

The city-level results show that vertical cartels have larger effects than horizontal cartels. I next examine whether these differences vary with firm size, as suggested by the theoretical mechanism: smaller retailers may have stronger incentives to deviate from horizontal coordination, while upstream participation can alter these incentives through wholesale terms.

The analysis uses the station-level sample obtained from the coarsened exact matching procedure described above. Each treated station is matched to at least one observationally similar station in the selected control market, and each treated station together with its matched controls defines a separate stack. I then estimate the following stacked difference-in-differences specification:

$$Y_{ist} = \alpha_{is} + \lambda_{st} + \beta_1 \text{Cartel}_{ist} + \beta_2 (\text{Cartel}_{ist} \times \text{Vertical}_s) + \beta_3 (\text{Cartel}_{ist} \times \text{Small}_i) + \beta_4 (\text{Cartel}_{ist} \times \text{Vertical}_s \times \text{Small}_i) + \varepsilon_{ist}, \quad (3)$$

where Cartel_{ist} equals one for a treated station while its cartel is active and zero otherwise. The specification includes station-by-stack fixed effects, α_{is} , and stack-by-week fixed effects, λ_{st} , so treated stations are compared with their matched controls within each stack.

Firm size is defined at the ownership-group level. Within each treated market, I aggregate gasoline storage capacity across stations sharing the same CNPJ root and classify groups in the bottom quartile of the resulting capacity distribution as small. Thus, Small_i captures the size of the retail firm rather than the capacity of an individual station.

Table 6 reports total effects by cartel structure and firm size in Panel A and the corresponding regression components in Panel B. To facilitate interpretation in terms of the effects of cartel operation, all reported coefficients are multiplied by -1 , so positive values indicate increases in the outcome while the cartel is active relative to the post-dissolution period. Retail-price effects are positive for both small and large firms and are larger under vertical cartels, but there is little evidence that they differ systematically by firm size. Con-

Table 6: Firm-Level Effects by Station Size

	(1) Retail Price	(2) Retail Markup	(3) Wholesale Markup
<i>Panel A. Total effects by cartel type and station size</i>			
Horizontal cartel, large station	0.0214*** (0.0018)	0.0185*** (0.0020)	0.0005 (0.0025)
Horizontal cartel, small station	0.0220*** (0.0024)	0.0181*** (0.0024)	0.0027 (0.0024)
Vertical cartel, large station	0.0387*** (0.0024)	0.0322*** (0.0026)	0.0010 (0.0033)
Vertical cartel, small station	0.0329*** (0.0041)	0.0198*** (0.0023)	0.0155*** (0.0047)
<i>Panel B. Individual components</i>			
Horizontal cartel effect, large stations	0.0214*** (0.0018)	0.0185*** (0.0020)	0.0005 (0.0025)
Small-station component in horizontal cartels	0.0006 (0.0030)	-0.0004 (0.0030)	0.0022 (0.0035)
Additional vertical-cartel component, large stations	0.0173*** (0.0031)	0.0136*** (0.0033)	0.0005 (0.0042)
Additional small-station component in vertical cartels	-0.0064 (0.0054)	-0.0120*** (0.0046)	0.0122** (0.0059)
Stack-Station Fixed Effects	Yes	Yes	Yes
Stack-Week Fixed Effects	Yes	Yes	Yes
Observations	98,369	98,369	98,369
Number of Stations	1,135	1,135	1,135

Note: This table reports estimates from stacked difference-in-differences regressions estimated on the matched firm-level sample. Because treatment is defined as the period after cartel dissolution, reported coefficients reverse the sign of the post-dissolution estimates and are interpreted as implied effects during cartel operation. Panel A reports total effects for each cartel type and station-size group. Panel B reports the individual contribution of each regression component to those total effects. A station is classified as small according to the definition described in the data section. All specifications include stack-by-station and stack-by-week fixed effects and are estimated using matching weights. Standard errors clustered at the station level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sistent with Table 5, vertical coordination therefore raises retail prices more than horizontal coordination without generating substantial size heterogeneity in the price effect.

The markup results display a different pattern. Under horizontal cartels, retail-markup effects are similar for small and large firms. Under vertical cartels, however, small firms receive smaller retail-markup gains: the estimated effect is 3.2 p.p. for large firms and 2.0 p.p. for small firms, implying that the small-firm differential is approximately one percentage point lower. At the same time, wholesale markups remain close to zero for large firms and for small firms in horizontal cartels, but increase by approximately 1.6 p.p. for small firms in vertical cartels.

This pattern is consistent with the documentary evidence discussed in Section 3. The case records indicate that smaller retailers were more likely to deviate from coordinated conduct and that, in vertical cartels, wholesalers could use differentiated wholesale prices and commercial terms to discipline these stations and keep them within the collusive arrangement. The station-level results are consistent with this mechanism: small stations face similar retail-price effects but capture a smaller increase in downstream margins, while wholesalers earn higher margins on sales to them.

Taken together, the evidence suggests that vertical coordination affects not only the overall magnitude of collusion but also the distribution of its gains across the vertical chain. Small retailers do not experience smaller retail-price effects, but they retain less of the associated retail-margin gain, consistent with wholesale terms being used to support a more inclusive downstream cartel.

5 Theoretical Model

I develop a model to examine whether the mechanisms documented in the cartel cases can rationalize the reduced-form results. The legal records indicate that smaller retailers are particularly prone to deviate from horizontal agreements and that, in vertical cartels,

wholesalers use differentiated wholesale terms to discipline downstream firms and sustain broader participation. The model formalizes these features in a capacity-constrained retail market and asks whether upstream participation can generate the larger retail-price and markup effects, together with the firm-size heterogeneity, documented empirically.

The model complements existing theories of vertical and hub-and-spoke collusion by emphasizing a different mechanism. Clark et al. (2024) study how retail shelf-space allocation creates asymmetries among upstream suppliers and affects their ability to collude. Van Cayseele and Miegielsen (2013) show how a common supplier can support downstream collusion through the threat of excluding deviating retailers. More recently, Gilo and Yehezkel (2025) study market-wide collusion between suppliers and retailers with differentiated products. In contrast, my model emphasizes heterogeneous downstream capacity and the use of discriminatory wholesale prices to alter retailers' participation incentives without excluding them from the market.

5.1 Setting

The economy consists of vertically related upstream wholesale and downstream retail markets. A homogeneous good is sold to final consumers with demand $D(p)$, which is twice continuously differentiable, strictly decreasing, and weakly concave in the retail price p .

The upstream market consists of N_w homogeneous wholesalers, each with constant marginal cost c and discount factor $\delta \in (0, 1)$. The downstream market consists of N_r retailers with discount factor $\theta \in (0, 1)$. Retailer i has storage capacity k_i , with total retail capacity

$$K = \sum_{i=1}^{N_r} k_i.$$

Retailers purchase the good at wholesale price p_w and sell it at retail price p_r , so p_w is their marginal cost.

As in Bos and Harrington (2010), capacity constraints play a central role. I impose two

assumptions.

Assumption 1. Let $p^M(p_w)$ denote the monopoly retail price given wholesale price p_w :

$$p^M(p_w) \in \arg \max_p \{ (p - p_w) D(p) \}.$$

For every retailer i ,

$$k_i < D(p^M(p_w)).$$

Thus, no individual retailer can serve monopoly demand.

Assumption 2. For every retailer i ,

$$\sum_{j \neq i} k_j \geq D(p_w).$$

Thus, after removing any one retailer, the remaining firms can jointly serve competitive demand.

Timing. The model has two stages.

1. **Cartel-formation stage.**

- (a) Retailers decide whether to form a downstream cartel. If so, they choose its membership Γ and aggregate capacity

$$K_\Gamma = \sum_{i \in \Gamma} k_i.$$

- (b) Wholesalers decide whether to form an upstream cartel.
- (c) If both cartels form, they decide whether to coordinate vertically. Without vertical coordination, the two cartels choose prices independently.

2. Repeated pricing stage.

Firms play an infinitely repeated pricing game conditional on the cartel structure chosen in the first stage.

- (a) **No vertical coordination.** Wholesalers first choose p_w , after which retailers choose p_r .
- (b) **Vertical coordination.** If the upstream and downstream cartels coordinate, wholesale and retail prices are jointly determined through Nash bargaining between the two cartels.

This timing nests five market structures summarized in Figure 3. I solve the model by backward induction. The next five subsections characterize equilibrium pricing under each market structure, while Subsection 5.7 derives the conditions under which these outcomes can be sustained in equilibrium.

5.2 Non-Cartel Benchmark

Wholesalers compete à la Cournot, so the wholesale price includes a Cournot markup over marginal cost:

$$p_w^N = c + \mu_w^N,$$

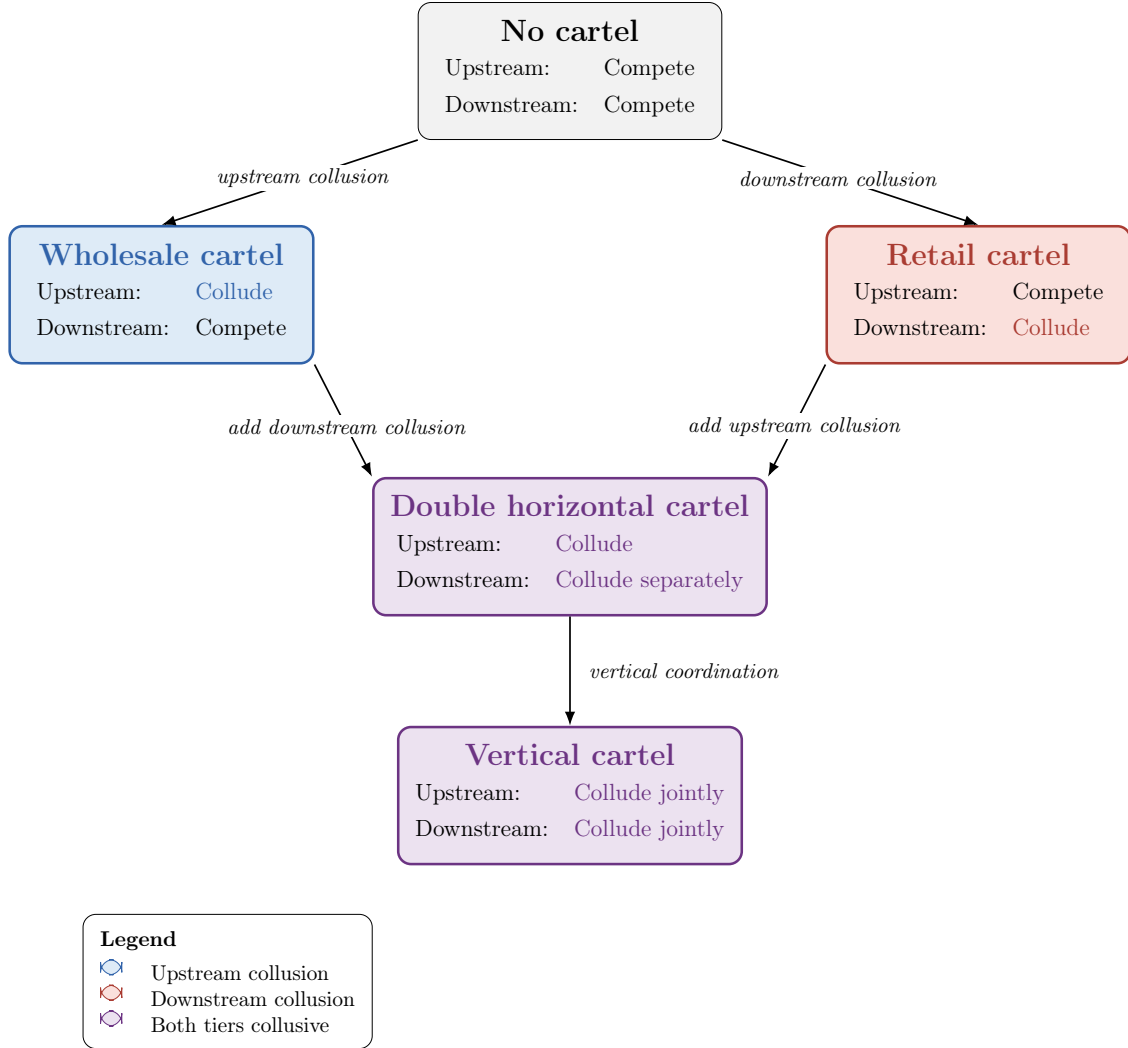
where $\mu_w^N \geq 0$ is decreasing in the number of wholesalers N_w .

Retailers compete in prices. Since the product is homogeneous and all retailers face the same wholesale price, Bertrand competition implies that retailers price at marginal cost:

$$p_r^N = p_w^N.$$

Thus, retailers earn zero variable profits, while wholesalers may earn positive profits due to upstream market power.

Figure 3: Cartel structures in the model.



5.3 Wholesale Cartel

Suppose wholesalers form an upstream cartel while retailers remain competitive. Bertrand competition downstream implies full pass-through of the wholesale price:

$$p_r = p_w.$$

The wholesale cartel therefore chooses

$$p_w^{WC} \in \arg \max_{p_w} \{(p_w - c)D(p_w)\},$$

Total cartel profit is

$$\Pi_w^{WC} = (p_w^{WC} - c) D(p_w^{WC}), \quad \Pi_r^{WC} = 0$$

Under symmetric profit sharing, each wholesaler receives Π_w^{WC}/N_w .

For the upstream cartel to form, wholesalers must prefer collusion to a one-period deviation. Under grim-trigger strategies, a deviating wholesaler slightly undercuts the cartel price and captures the entire wholesale market for one period, after which wholesalers revert permanently to Bertrand competition. The no-deviation condition is therefore

$$\frac{1}{1-\delta} \frac{\Pi_w^{WC}}{N_w} \geq \Pi_w^{WC},$$

or, equivalently,

$$\delta \geq 1 - \frac{1}{N_w}. \quad (4)$$

5.4 Retail Cartel

Following Bos and Harrington (2010), suppose retailers form a capacity-constrained downstream cartel with total capacity $K_\Gamma = \sum_{i \in \Gamma} k_i$, where Γ denotes the set of cartel members, while wholesalers remain non-cooperative. Retailers outside the cartel form a competitive fringe. They slightly undercut the cartel price and, in the limit as the price difference converges to zero, sell up to capacity, so outsider i earns $(p_r - p_w)k_i$.

The cartel faces residual demand $D(p_r) - (K - K_\Gamma)$, where $K - K_\Gamma$ is total fringe capacity, and shares profits proportionally to capacity. Thus, insider i earns $[D(p_r) - (K - K_\Gamma)](p_r - p_w)k_i/K_\Gamma$. For an insider not to deviate, its discounted collusive payoff must exceed the one-period gain from undercutting the cartel and selling its full capacity. The Bos–Harrington

incentive constraint is therefore

$$\frac{1}{1-\theta} [D(p_r) - (K - K_\Gamma)] (p_r - p_w) \frac{k_i}{K_\Gamma} \geq (p_r - p_w) k_i,$$

which simplifies to

$$D(p_r) \geq K - \theta K_\Gamma. \quad (5)$$

Since demand is decreasing, the highest sustainable retail price is $\bar{p}(K_\Gamma) = D^{-1}(K - \theta K_\Gamma)$. Greater cartel capacity therefore relaxes the incentive constraint and allows a weakly higher price.

Conditional on K_Γ , the cartel maximizes residual-demand profits subject to (5). Its pricing rule is

$$p_r^{RC}(p_w, K_\Gamma) = \min \{p^*(p_w, K_\Gamma), \bar{p}(K_\Gamma)\}, \quad (6)$$

where the unconstrained residual-monopoly price $p^* = p^*(p_w, K_\Gamma)$ satisfies

$$D(p^*) - (K - K_\Gamma) + (p^* - p_w) D'(p^*) = 0. \quad (7)$$

Wholesale pricing. Wholesalers anticipate the retail cartel's pricing response when choosing quantities. If (5) binds, the retail price is pinned down at $\bar{p}(K_\Gamma)$ and therefore no longer responds to the wholesale price. Final demand is consequently fixed at $D(\bar{p}(K_\Gamma))$. In this region, a higher wholesale price does not reduce quantity, so wholesalers have an incentive to raise p_w up to the retail price, $p_w = \bar{p}(K_\Gamma)$, thereby extracting the entire downstream margin. The retail cartel then earns zero profit. Such a configuration cannot be preferred whenever there exists an interior configuration in which the retail cartel earns strictly positive profits. I therefore focus on positive-profit equilibria in which the downstream incentive constraint is slack and the retail price responds to the wholesale price. The Online Appendix C formally characterizes the boundary case.

In this interior region, let $Q = D(p_r^{RC})$ and $P(Q) = D^{-1}(Q)$. The retail cartel's first-

order condition implies the derived inverse demand faced by wholesalers,

$$P_w^{RC}(Q, K_\Gamma) = P(Q) + [Q - (K - K_\Gamma)] P'(Q). \quad (8)$$

Wholesalers compete à la Cournot over this derived demand. The symmetric equilibrium quantity Q^{RC} satisfies

$$P_w^{RC}(Q^{RC}, K_\Gamma) - c + \frac{Q^{RC}}{N_w} \frac{\partial P_w^{RC}(Q^{RC}, K_\Gamma)}{\partial Q} = 0, \quad (9)$$

which implies the equilibrium wholesale price

$$p_w^{RC} = c - \frac{Q^{RC}}{N_w} \frac{\partial P_w^{RC}(Q^{RC}, K_\Gamma)}{\partial Q}. \quad (10)$$

Thus, the retail-cartel structure combines upstream market power with downstream collusion. Nevertheless, the retail price need not exceed the vertically integrated monopoly price because the competitive fringe limits the residual demand internalized by the cartel.

Theorem 1. Let $p^M(c) \in \arg \max_p \{(p - c)D(p)\}$ denote the vertically integrated monopoly price. Consider a non-fully-inclusive retail cartel and suppose residual-demand profits are single-peaked. If

$$K - K_\Gamma \geq (p_w^{RC} - c) [-D'(p^M(c))], \quad (11)$$

then $p_r^{RC} \leq p^M(c)$.

The proof is provided in the Online Appendix C. The condition compares the downward pressure from fringe capacity with the upward pressure generated by the upstream markup. When fringe capacity is sufficiently large, the former dominates and the retail-cartel price remains weakly below the integrated monopoly price.

5.5 Double Horizontal Cartel

Suppose wholesalers and retailers both form cartels but do not coordinate vertically. The retail cartel follows the same pricing strategy as in the previous subsection, so for any wholesale price p_w , $p_r^{DH}(p_w, K_\Gamma) = \min \{p^*(p_w, K_\Gamma), \bar{p}(K_\Gamma)\}$. As before, a configuration in which the downstream incentive constraint binds leaves the retail cartel with zero profit under the boundary refinement and is not selected whenever a positive-profit interior configuration exists. I therefore focus on the interior region.

The difference from the retail-cartel case is upstream pricing. Rather than competing à la Cournot, wholesalers form a cartel and choose the wholesale price anticipating the retail cartel's response:

$$p_w^{DH} \in \arg \max_{p_w} \{(p_w - c)D(p^*(p_w, K_\Gamma))\}. \quad (12)$$

The resulting retail price is $p_r^{DH} = p^*(p_w^{DH}, K_\Gamma)$, subject to the Bos–Harrington price cap. Thus, double-horizontal collusion generates two independent markups, while fringe capacity continues to limit the residual demand internalized by the retail cartel.

Theorem 2. Let $p^M(c)$ denote the vertically integrated monopoly price and suppose residual-demand profits are single-peaked. If

$$K - K_\Gamma \geq (p_w^{DH} - c) [-D'(p^M(c))], \quad (13)$$

then $p_r^{DH} \leq p^M(c)$. Conversely, if the inequality is reversed and $\bar{p}(K_\Gamma) \geq p^M(c)$, then $p_r^{DH} \geq p^M(c)$, with strict inequality whenever $\bar{p}(K_\Gamma) > p^M(c)$.

The proof is provided in the Online Appendix C. Relative to Theorem 1, the key difference is that the relevant wholesale price is now chosen by an upstream cartel rather than by Cournot competitors. Because the wholesale cartel generally has greater upstream market power, $p_w^{DH} - c$ can be larger than $p_w^{RC} - c$, making the condition for the retail price to remain below the integrated monopoly price more demanding. Thus, while fringe capacity continues

to constrain downstream pricing in both structures, the additional upstream market power under double-horizontal collusion makes prices above $p^M(c)$ more likely.

Although the price comparison is ambiguous, the comparison in total industry profits is not.

Theorem 3. Suppose $(p - c)D(p)$ is strictly concave and let $\Pi^M = (p^M(c) - c)D(p^M(c))$ denote integrated monopoly profit. If Π_w^{DH} and Π_R^{DH} denote aggregate upstream and downstream profits under double-horizontal collusion, then

$$\Pi_w^{DH} + \Pi_R^{DH} \leq \Pi^M, \quad (14)$$

with strict inequality whenever $p_r^{DH} \neq p^M(c)$. Moreover, there exists a division of Π^M between wholesalers and retailers that makes both tiers weakly better off than under the double-horizontal cartel.

The proof is provided in the Online Appendix C. The result follows because wholesale prices only divide surplus across tiers, while total industry profit is $(p_r^{DH} - c)D(p_r^{DH})$, which is maximized at $p^M(c)$. Hence, whenever double-horizontal collusion does not implement the integrated monopoly price, vertical coordination creates additional surplus that can be divided between wholesalers and retailers.

5.6 Vertical Cartel

Suppose wholesalers and retailers coordinate vertically. Motivated by the documentary evidence in Section 3, vertical coordination has two features. First, wholesalers can offer different wholesale terms to cartel participants and non-participants, thereby affecting retailers' incentives to participate in the downstream agreement. Second, the two tiers jointly determine retail and wholesale prices

Vertical discrimination. Let p_w^I denote the wholesale price charged to cartel insiders and p_w^O the price charged to outsiders, with $p_w^O > p_w^I \geq c$. For an outsider to remain active and undercut the cartel price by an arbitrarily small amount $\xi > 0$, it must be that

$$p_w^O \leq p_r^V - \xi. \quad (15)$$

Hence, as $\xi \rightarrow 0$, the highest feasible outsider wholesale price is $p_w^O = p_r^V$. Vertical discrimination therefore weakens the outside option available to retailers relative to a horizontal retail cartel.

Proposition 1 (Vertical discrimination and full participation). Under vertical collusion, the optimal outsider wholesale price attains the feasibility bound in (15). As $\xi \rightarrow 0$, $p_w^O = p_r^V$. At this price, all retailers prefer to participate in the cartel, the competitive fringe is eliminated, and the vertical cartel charges the integrated monopoly price,

$$p_r^V = p^M(c) \in \arg \max_p \{(p - c)D(p)\}. \quad (16)$$

The proof is provided in the Online Appendix C. The intuition follows from the effect of the outsider wholesale price on cartel participation. A higher p_w^O reduces the profit available to a retailer that remains outside the agreement, making cartel participation more attractive. Greater downstream participation, in turn, reduces fringe capacity and allows the cartel to internalize a larger share of market demand and generate greater joint surplus. The vertical cartel therefore sets p_w^O as high as feasibility permits. In the limit, $p_w^O = p_r^V$, so an outsider earns zero margin, whereas an insider pays $p_w^I < p_r^V$ and receives a positive share of cartel surplus. All retailers consequently join the cartel. With the competitive fringe eliminated, the vertical cartel internalizes the entire margin between the retail price and marginal cost c and therefore chooses the integrated monopoly price.

Vertical coordination thus determines the size of the surplus available to cartel partici-

pants. What remains is to determine how this surplus is divided between the upstream and downstream tiers, which is governed by the insider wholesale price p_w^I .

Vertical bargaining. Let $\Pi^M = (p^M(c) - c)D(p^M(c))$ denote total monopoly profit. Because vertical coordination fixes the retail price at $p^M(c)$, the insider wholesale price does not affect total surplus but determines how it is divided between the two tiers. In particular, $\Pi_w^V(p_w^I) = D(p^M(c))(p_w^I - c)$ and $\Pi_r^V(p_w^I) = D(p^M(c))(p^M(c) - p_w^I)$. I model the determination of p_w^I as Nash bargaining between the upstream and downstream cartels, consistent with the documentary evidence that wholesalers and retailers jointly discussed prices, wholesale terms, and profit margins. Nash bargaining therefore provides a parsimonious way to represent how the monopoly surplus is allocated between the two tiers as a function of their bargaining power and outside options.

The relevant outside options are the outcomes that can arise if vertical coordination fails while downstream collusion remains possible. Importantly, without vertical coordination wholesalers can no longer use discriminatory wholesale terms to induce full downstream participation. The retail cartel therefore reverts to the non-fully-inclusive structure characterized above, in which some retailers may remain in the competitive fringe. Wholesalers may then either remain non-cooperative, generating the retail-cartel outcome, or form an independent upstream cartel, generating the double-horizontal outcome whenever such a cartel is sustainable and individually rational. Let $\mathcal{O} = \{RC, DH\}$ denote this set of feasible no-vertical structures. Because the vertical agreement requires wholesaler participation, the disagreement regime is the feasible alternative that delivers the highest upstream payoff, $o^* \in \arg \max_{o \in \mathcal{O}} \Pi_w^o$. The associated disagreement payoffs are $(\Pi_r^0, \Pi_w^0) = (\Pi_r^{o^*}, \Pi_w^{o^*})$, so both tiers' outside options are evaluated under the same fallback market structure.

The insider wholesale price is determined through Nash bargaining between the upstream and downstream cartels, subject to the incentive constraints required to sustain

vertical coordination:

$$\begin{aligned}
& \max_{p_w^I} [D(p^M(c))(p_w^I - c) - \Pi_w^0]^{1-\lambda} [D(p^M(c))(p^M(c) - p_w^I) - \Pi_r^0]^\lambda \\
& \text{s.t.} \quad \theta \geq \max \left\{ 1 - \frac{D(p^M(c))}{K}, \frac{\Pi_r^D(p_w^I) - \Pi_r^V}{\Pi_r^D(p_w^I) - \Pi_r^0} \right\}, \\
& \quad \delta \geq \max \left\{ 1 - \frac{1}{N_w}, \frac{\Pi^M - \Pi_w^V}{\Pi^M - \Pi_w^0} \right\}.
\end{aligned} \tag{17}$$

Here, $\lambda \in (0, 1)$ is the bargaining weight of the retail cartel, and $\Pi_r^D(p_w^I) = \max_{p_r} D(p_r)(p_r - p_w^I)$ denotes the retail cartel's optimal one-period deviation payoff.

The first constraint collects the two conditions required on the downstream side. The term $1 - D(p^M(c))/K$ is the equation (5) evaluated at full participation and the integrated monopoly price, ensuring that retailers are sufficiently patient to sustain the coordinated retail price. The second term ensures that the retail cartel does not prefer a one-period deviation from the vertical agreement, after which it receives its disagreement payoff Π_r^0 .

The second constraint imposes the corresponding requirements on wholesalers. The first is equation (4) required for the upstream cartel itself to be sustainable. The second term ensures that wholesalers do not deviate from the negotiated vertical agreement by extracting the full monopoly surplus for one period and subsequently reverting to their disagreement payoff Π_w^0 .

The Online Appendix C details the derivation of these conditions. When none of the incentive constraints binds, Nash bargaining yields

$$p_w^{I,V} = c + \frac{\Pi_w^0 + (1 - \lambda)(\Pi^M - \Pi_w^0 - \Pi_r^0)}{D(p^M(c))}. \tag{18}$$

Equivalently, the incremental surplus from vertical coordination is divided according to

bargaining weights,

$$\Pi_w^V = \Pi_w^0 + (1 - \lambda) (\Pi^M - \Pi_w^0 - \Pi_r^0), \quad \Pi_r^V = \Pi_r^0 + \lambda (\Pi^M - \Pi_w^0 - \Pi_r^0).$$

Thus, $p^M(c)$ determines total cartel surplus, while $p_w^{I,V}$ determines its division between wholesalers and retailers.

These predictions rationalize the main reduced-form results. First, discriminatory wholesale pricing weakens the outside option of retailers that would otherwise remain in the competitive fringe. By inducing broader downstream participation, vertical coordination eliminates the fringe constraint that limits horizontal retail cartels and allows the cartel to sustain the integrated monopoly price. This mechanism is consistent with the larger retail-price effects observed for vertical cartels and with the station-level evidence that smaller retailers, which have stronger incentives to deviate, retain a smaller share of the collusive surplus.

Second, once full participation is achieved, wholesalers and retailers jointly negotiate the division of the additional surplus generated by vertical coordination. The Nash bargaining solution captures this joint determination through the insider wholesale price, which allocates surplus between the two tiers without changing the retail price. Consequently, the model does not imply that vertical coordination must raise the wholesale price. The negotiated price depends on wholesalers' bargaining leverage and, in particular, on their disagreement payoff. When the wholesalers' outside option is weak, they may accept a lower insider wholesale price because the additional surplus generated by vertical coordination can still leave them better off than under the no-vertical outcome. Hence, vertical coordination can substantially increase retail prices and retail markups while producing a smaller or even negative change in wholesale prices, consistent with the weaker reduced-form evidence for wholesale-price effects.

5.7 Cartel Formation

Having characterized pricing under each market structure, I now determine which cartel configurations can arise in equilibrium. The pricing game may support multiple cartel compositions, so I first define stability and then use the cartel-formation stage to select among stable configurations.

Definition 1 (Stability). A cartel configuration is stable if all participating firms prefer to remain in the agreement and firms outside the cartel prefer to remain outside. In the present setting, this requires:

1. **Downstream stability.** The smallest retailer inside the cartel prefers to remain an insider, while the largest retailer outside prefers to remain in the competitive fringe.
2. **Upstream stability.** Whenever wholesalers collude, they prefer the cartel outcome to the relevant alternative without upstream collusion and have no profitable unilateral deviation from the upstream agreement.

The first condition follows the stability concept in Bos and Harrington (2010). Because retailers' participation incentives are monotone in capacity, it is sufficient to check the smallest insider and the largest outsider to characterize the stability of the downstream cartel. The second condition extends this concept to the upstream tier. Wholesalers must both benefit from forming the cartel relative to the relevant non-collusive alternative and be sufficiently patient to sustain the agreement.

The stability argument has an especially sharp implication for vertical cartels. Proposition 1 shows that discriminatory wholesale pricing reduces the payoff from remaining outside the agreement. At the optimal outsider price, $p_w^O \rightarrow p_r^V$, an outsider earns zero variable profit, while an insider facing $p_w^I < p_r^V$ receives a positive share of cartel surplus. Hence, any retailer left outside a positive-profit vertical cartel would prefer to join. The only stable vertical-cartel composition is therefore the fully inclusive one, $\Gamma = \{1, \dots, N_r\}$.

This argument also rules out the boundary outcome discussed in the retail-cartel pricing problem. When the downstream incentive constraint binds and the wholesale price reaches the retail-price cap, wholesalers extract the entire downstream margin and retailers earn zero profit. Retailers are then indifferent between participating and remaining outside. I exclude this knife-edge outcome by requiring members of an implemented cartel to strictly prefer participation to their outside option. The boundary configuration is therefore not stable whenever a positive-profit interior configuration is available.

The pricing game may admit multiple stable downstream cartel compositions. The formation stage therefore acts as an equilibrium-selection mechanism. Because retailers move first, they anticipate the wholesalers' equilibrium response to each feasible cartel composition, including whether wholesalers remain non-cooperative, form an independent upstream cartel, or coordinate vertically when feasible. Retailers then choose, among all stable configurations, the cartel membership that maximizes their resulting equilibrium profit. Formally, letting $\mathcal{F}$ denote the set of stable downstream configurations and $u(\Gamma)$ the equilibrium upstream response to cartel composition Γ , the selected cartel satisfies

$$\Gamma^* \in \arg \max_{\Gamma \in \mathcal{F}} \Pi_r^{u(\Gamma)}(\Gamma).$$

Thus, cartel formation selects among the multiple stable outcomes of the pricing game by allowing retailers to internalize the wholesalers' subsequent response when choosing cartel size.

The Appendix complements these analytical results with a calibrated numerical example that illustrates the formation mechanism and evaluates whether the model can reproduce the empirical ranking of horizontal and vertical cartel effects.

5.8 Model Implications

The model formalizes the mechanisms documented in the cartel cases and provides a rationale for the reduced-form differences between horizontal and vertical cartels. In the legal records, horizontal cartels frequently struggle with retailers that undercut the coordinated price, particularly smaller stations. In the model, these firms form the competitive fringe. Their capacity reduces the residual demand internalized by cartel members and limits the price that a horizontal retail cartel can sustain. This mechanism is consistent with the reduced-form evidence showing positive but relatively smaller effects of horizontal cartels on retail prices and markups.

Vertical participation changes this constraint through the wholesale relationship. The documentary evidence indicates that wholesalers could offer less favorable wholesale terms to retailers that remained outside or deviated from the agreement. In the model, this discrimination weakens the outside option of non-participating retailers and induces full downstream participation. By eliminating the competitive fringe, the vertical cartel internalizes the full vertical margin and can implement the integrated monopoly price. This mechanism rationalizes the substantially larger retail-price effects observed for vertical cartels. It is also consistent with the station-level evidence: smaller retailers experience similar retail-price effects but smaller retail-markup gains under vertical coordination, while wholesalers earn higher margins on sales to these firms.

The model also explains why the evidence on wholesale prices and markups is weaker. Once full participation is achieved, wholesalers and retailers jointly negotiate the division of cartel surplus through the insider wholesale price. This price therefore depends on bargaining leverage and disagreement payoffs rather than solely on the degree of market power. Wholesalers with a weak outside option may accept a relatively low insider wholesale price because vertical coordination generates additional joint surplus and can still leave them better off than under the no-vertical alternative. Consequently, the model can generate large

increases in retail prices and retail markups without requiring an equally large increase in wholesale prices, consistent with the reduced-form evidence.

6 Conclusion

This paper studies how vertical participation changes the organization and consequences of collusion in gasoline retail markets. Combining documentary evidence from Brazilian antitrust cases, reduced-form estimates, and a model of capacity-constrained downstream collusion with upstream participation, I show that vertical cartels are not simply horizontal cartels with an additional upstream layer. Wholesaler participation changes both retailers' incentives to join the agreement and the way collusive surplus is divided across the supply chain.

The empirical results show that vertical cartels generate substantially larger increases in retail prices and retail markups than horizontal cartels. The evidence on wholesale prices and wholesale markups is weaker but broadly consistent with upstream firms benefiting from vertical coordination. Station-level heterogeneity provides additional evidence on the mechanism. Small retailers experience similar retail-price effects but smaller retail-markup gains under vertical cartels, while wholesalers earn higher margins on sales to these firms.

The model rationalizes these patterns using the mechanisms documented in the legal records. Under horizontal collusion, smaller retailers may remain outside the agreement, forming a competitive fringe that limits the residual demand internalized by cartel members and constrains the sustainable retail price. Under vertical coordination, wholesalers can use discriminatory wholesale terms to weaken the outside option of non-participating retailers and induce full downstream participation. Eliminating the fringe allows the cartel to internalize the full vertical margin and sustain a higher retail price. Wholesalers and retailers then bargain over the division of the resulting surplus through the insider wholesale price. Consequently, vertical coordination can generate large retail-price and retail-markup

effects without necessarily producing an equally large increase in wholesale prices. A numerical example and calibrated exercise further show that these mechanisms can reproduce the qualitative ranking observed in the data.

These findings have implications for antitrust enforcement in vertically related markets. Vertical relationships are often associated with efficiencies such as improved coordination or reduced double marginalization, but the same instruments can strengthen collusion when upstream firms participate in the agreement. Wholesale terms can be used to discipline deviating retailers, reduce fringe competition, and sustain broader cartel participation. Evaluating cartel harm therefore requires attention not only to whether firms collude, but also to how the cartel is organized across the supply chain. In the Brazilian gasoline market, this distinction helps explain why vertical cartels generate substantially larger consumer harm than purely horizontal retail cartels.

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A Appendix: Numerical Example and Calibration

I complement the reduced-form evidence with a calibrated numerical exercise based on the theoretical model. The exercise is not a structural estimation and is not intended to quantify cartel damages. Instead, it asks whether a transparent parameterization disciplined by observable market characteristics can reproduce the qualitative ranking documented empirically.

The calibration reproduces the main pattern in the data. Vertical coordination generates larger effects on retail prices and retail markups than a horizontal retail cartel, together with larger wholesale-markup effects. The exercise therefore provides a quantitative illustration of the model’s mechanism, in which horizontal cartels are constrained by fringe capacity and participation incentives, while vertical coordination weakens retailers’ outside options and changes the division of surplus across tiers.

I focus on qualitative comparisons because the available data do not support a full structural estimation. Station-level quantities are not observed, preventing the estimation of case-specific demand and market shares, and the actual membership of each cartel is not directly observed. The model also abstracts from spatial differentiation and other forms of firm and demand heterogeneity. Accordingly, the calibration is intended to assess whether the model can reproduce the ranking of outcomes across cartel structures rather than the exact magnitude of the reduced-form estimates.

Calibration Strategy

I normalize the non-cartel retail price and quantity to $p_0 = Q_0 = 1$ and assume linear demand, $D(p) = A - Bp$. Given the benchmark market-demand elasticity ε_0 , the normalization implies $B = -\varepsilon_0$ and $A = 1 - \varepsilon_0$. In the baseline, I set $\varepsilon_0 = -0.399$ following Santos (2013), interpreting this parameter as the aggregate market-demand elasticity rather than the residual elasticity faced by an individual station.

I discipline market structure using the Brasília/DF cartel case (Case 8). I preserve the empirical distribution of station-level gasoline storage capacity and rescale aggregate retail capacity to $K = 1.10Q_0$, introducing moderate excess capacity that allows the competitive fringe to constrain downstream collusion. I set $N_w = 3$, consistent with Chaves and Duarte (2025), who identify three wholesalers as accounting for the main upstream activity in this market.

Marginal cost is recovered from the non-cartel benchmark. Because retailers are competitive and wholesalers compete à la Cournot, $p_r^N = p_w^N = p_0$. The symmetric Cournot first-order condition implies

$$c = p_0 + \frac{Q_0}{N_w} P'(Q_0), \quad (\text{A1})$$

where $P(Q) = D^{-1}(Q)$. The integrated monopoly price is then $p^M = (A/B + c)/2$.

For horizontal retail cartels, retailers are ordered from largest to smallest by capacity, and a candidate cartel containing the m largest firms is considered for each $m = 1, \dots, N_r - 1$. For every candidate, I solve the pricing game and impose the stability conditions derived in the model: the smallest insider prefers to remain inside, the largest outsider prefers to remain outside, and the equilibrium lies in the positive-profit interior region. Among stable configurations, retailers select the cartel that maximizes their aggregate profit, anticipating the corresponding upstream response.

I set the retailer and wholesaler discount factors to $\theta = \delta = 0.95$, representing sufficiently patient firms while allowing the incentive constraints to remain operative. I solve the model under the five structures characterized above: no cartel, wholesale cartel, retail cartel, double-horizontal cartel, and vertical cartel. Under vertical coordination, all retailers participate and the retail price equals the integrated monopoly price, $p_r^V = p^M(c)$. The insider wholesale price is determined through Nash bargaining subject to the participation and deviation constraints. Rather than imposing a bargaining weight, I compute the set of feasible $\lambda \in [0, 1]$ and use its median for the reported vertical-cartel outcomes.

Finally, when solving the cartel-formation stage, I compare only stable and feasible structures. Retailers choose the cartel configuration anticipating the wholesalers’ equilibrium response, as in the formation game.

Baseline Counterfactuals

Table A1 reports the profit implications of the baseline calibration.

Table A1: Cartel Participation and Profits

Outcome	Market structure			
	WC	RC	DH	VC
<i>Cartel participation</i>				
Share of station storage capacity in cartel (%)	–	30.1	64.1	100.0
<i>Profit allocation</i>				
Wholesale profit change (%)	33.3	24.3	12.0	28.1
Retail profit (% of benchmark producer profit)	0.0	0.1	0.4	5.2
Total producer profit change (%)	33.3	24.4	12.4	33.3

Note: WC denotes wholesale cartel, RC retail cartel, DH double-horizontal cartel, and VC vertical cartel. The participation row is the sum of member stations’ storage capacities divided by aggregate station storage capacity. RC and DH sizes are selected endogenously from the empirical capacity distribution. All vertical outcomes use the median feasible bargaining weight. Profit changes are relative to the non-cartel benchmark; retail profit is scaled by benchmark producer profit.

When only wholesalers collude, wholesale and total producer profits increase by 33% relative to the benchmark, while retailers earn zero variable profit. When only retailers collude, the endogenous retail cartel includes 30% of total retail capacity. If wholesalers also form an independent cartel, downstream participation rises to 74.6%. Thus, upstream conduct affects the profitability of retail participation and, consequently, cartel composition. At the same time, the double-horizontal structure illustrates the inefficiency created by separate upstream and downstream pricing where retail profits increase, but wholesale profits fall relative to the benchmark as the downstream markup reduces final demand.

Vertical coordination eliminates this conflict by maximizing joint surplus and dividing

it through the insider wholesale price. In the baseline, all retailers participate and total producer profit exceeds that under the retail-horizontal and double-horizontal structures. The exercise therefore illustrates that vertical coordination can generate additional surplus by eliminating the competitive fringe and avoiding the inefficiencies associated with independent pricing across tiers.

Table A2 reports the corresponding price and markup effects.

Table A2: Cartel Price and Markup Effects

Outcome	Market structure			
	WC	RC	DH	VC
<i>Prices</i>				
Retail price change (%)	83.5	53.0	142.3	83.5
Wholesale price change (%)	83.5	48.2	133.0	77.0
<i>Markup allocation</i>				
Wholesale markup change (%)	100.0	57.7	159.2	92.2
Retail markup (% of benchmark total markup)	0.0	5.8	11.2	7.8
Total markup change (%)	100.0	63.5	170.4	100.0

Note: Price changes are relative to the non-cartel benchmark. Wholesale and total markup changes use their benchmark values; retail markup is scaled by benchmark total markup. VC is evaluated at the median feasible bargaining weight.

The calibrated magnitudes are substantially larger than the reduced-form estimates and should not be interpreted as quantitative estimates of cartel overcharges. The relevant comparison is the ranking across structures. Relative to the retail-horizontal cartel, vertical coordination produces larger retail-price, retail-markup, and wholesale-markup effects, consistent with the reduced-form evidence. While the retail-horizontal cartel remains constrained by fringe capacity, vertical coordination eliminates the fringe and implements the integrated monopoly price.

The double-horizontal cartel generates the largest price and markup effects. This outcome is best viewed as a counterfactual benchmark rather than the empirical counterpart of the observed vertical cases. Because upstream and downstream cartels set prices inde-

pendently, the structure combines two collusive margins and can generate severe double marginalization, raising final prices while producing less total surplus than coordinated vertical collusion.

Welfare and Rent Sharing

Table A3 reports the welfare effects of the alternative cartel structures.

Table A3: Cartel Welfare Effects

Outcome	Market structure			
	WC	RC	DH	VC
Consumer surplus change (%)	-55.6	-37.8	-81.3	-55.6
Producer surplus change (%)	33.3	24.4	12.4	33.3
Total welfare change (%)	-20.0	-12.9	-43.8	-20.0

Note: Consumer surplus, producer surplus, and total welfare are percentage changes relative to the non-cartel benchmark. VC is evaluated at the median feasible bargaining weight.

All cartel configurations reduce consumer surplus and total welfare relative to the non-cartel benchmark. The retail cartel generates the smallest welfare loss because incomplete downstream participation limits the increase in the retail price. The wholesale and vertical cartels generate larger consumer losses because both implement the integrated monopoly price in the baseline calibration. The double-horizontal cartel produces the largest welfare loss, as independent upstream and downstream pricing generates severe double marginalization and the largest retail-price distortion.

Table A4 reports how monopoly profit is divided under vertical coordination across the feasible range of retailer bargaining weights. Although the feasible range of λ is relatively wide, the implied allocation of monopoly profit remains concentrated upstream. Retailers receive between 1.4 and 5.9 percent of monopoly profit, while wholesalers capture the remainder. This asymmetry reflects the wholesalers' strong bargaining position in the calibrated model. Importantly, the small retail share does not imply that retailers are worse off under

vertical coordination. The bargaining solution first guarantees each tier its no-vertical disagreement payoff and then divides the incremental surplus created by vertical coordination. Retailers can therefore prefer vertical coordination even when wholesalers capture most of the resulting monopoly profit.

Table A4: Division of Monopoly Profit under Vertical Coordination

Outcome	Feasible bargaining weight		
	Lower bound	Median	Upper bound
Retailer bargaining weight, λ	0.1916	0.5704	0.9493
Retailer share of monopoly profit (%)	1.4	3.9	6.4
Wholesaler share of monopoly profit (%)	98.6	96.1	93.6

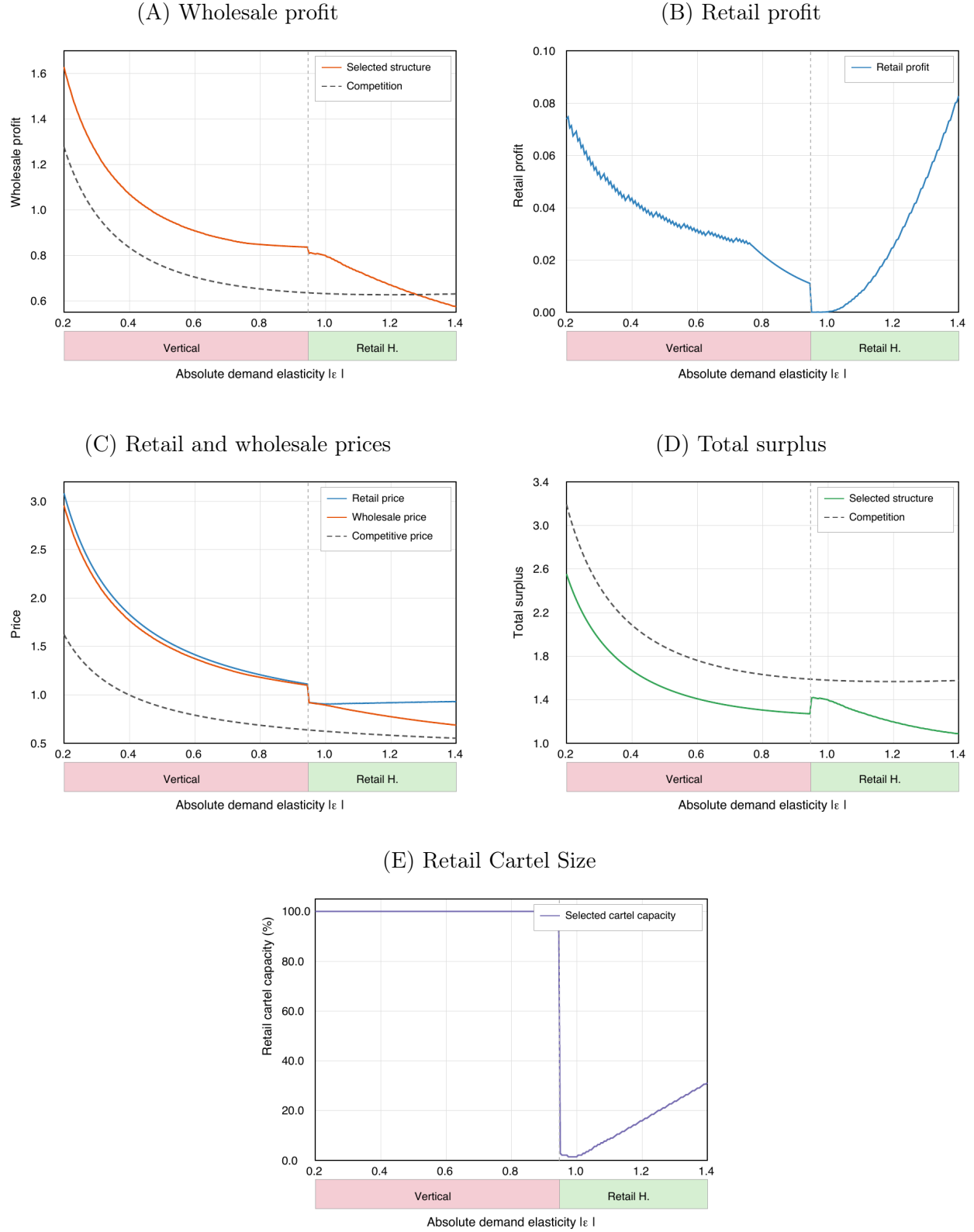
Note: This is the only table that reports the lower, median, and upper feasible bargaining weights. Tier profit shares are percentages of total monopoly profit and sum to 100 percent within each column.

Sensitivity to Demand Elasticity

I examine how cartel formation changes with market-demand elasticity while holding the baseline market structure fixed. For each elasticity value, I recalibrate the linear demand curve to preserve the normalization $p_0 = Q_0 = 1$, keep marginal cost and the empirical capacity distribution fixed, solve all feasible cartel structures, and select the retailer-preferred outcome subject to wholesaler participation.

Figure A1 shows that demand elasticity affects both the cartel structure selected by the model and the size of the downstream cartel. When demand is relatively inelastic, the vertical cartel is selected and all retail capacity participates. As demand becomes more elastic, the quantity loss from integrated monopoly pricing becomes more costly, reducing the gains from vertical coordination. Around $|\varepsilon| = 0.95$, the model switches to a retail-horizontal cartel and participation falls sharply. Within the retail-cartel region, however, the optimal cartel expands as elasticity rises because the competitive fringe becomes more important relative to total demand, increasing the value of internalizing additional downstream capacity.

Figure A1: Sensitivity of Cartel Outcomes to Demand Elasticity



Note: The figure reports model-selected outcomes as the absolute value of market-demand elasticity varies. For each value, the linear demand curve is recalibrated to preserve $p_0 = Q_0 = 1$, while the remaining baseline market characteristics are held fixed. The selected structure maximizes retailer profit among feasible outcomes subject to wholesaler participation. Pink denotes a vertical cartel and green a retail cartel. The dashed line marks the switch between the two structures. Vertical-cartel outcomes use the median feasible bargaining weight.

The price and profit panels show how this changes the division of surplus across tiers. At the transition, both retail and wholesale prices fall because the retail cartel remains constrained by the fringe. Beyond the transition, the retail price is roughly stable while the wholesale price declines as demand becomes more elastic. Since higher retail prices generate larger quantity losses, wholesalers reduce their price to support downstream sales. This widens the retail-wholesale margin, lowers wholesale profits, and raises retail profits as the cartel expands.

The total-surplus panel shows a discrete increase at the transition because the retail price falls and the market moves away from integrated monopoly pricing. Total surplus then declines as the retail cartel expands and maintains a positive retail markup. Overall, the exercise shows that relatively inelastic demand favors full vertical coordination, whereas more elastic demand favors a partially inclusive retail cartel whose size increases with demand elasticity.